

Semantic Segmentation Benchmarking and Self- Supervised Label-Efficiency Study for Breast Ultrasound Lesion Segmentation

CSE 438 | Digital Image Processing | Section 04

Group F

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Dataset: BCSD-2024 Breast Ultrasound Lesion Segmentation

Abstract

This project evaluates supervised and self-supervised semantic segmentation strategies for breast ultrasound lesion segmentation using the BCSD-2024 dataset. Part A of the project established a patient-level benchmark among DeepLabV3-ResNet50, SegFormer-B0, and YOLOv26-Sem using a common split. The dataset contains 1,521 image-mask pairs from 10 patients and was partitioned into 1,035 training, 318 validation, and 168 test images with no patient overlap. SegFormer-B0 achieved the strongest fully supervised test result with mIoU 0.8758, lesion IoU 0.7579, and mean Dice 0.9296. Part B of the project reused the identical split, treating the 1,035-image training set as unlabelled SSL data and the 318-image validation set as labelled downstream data. SimCLR, BYOL, MAE, and DINOv2 were each pretrained for 50 epochs and attached to the SegFormer-style decoder before 50-epoch supervised fine-tuning. DINOv2 achieved the best SSL result with an mIoU of 0.7633, recovering 87.2% of the Part A baseline while using only 30.7% of the labels. SimCLR and BYOL recovered 82.6% and 81.1%, whereas MAE recovered 55.9%.

Keywords—semantic segmentation, breast ultrasound, SegFormer, DeepLabV3, YOLOv26-Sem, self-supervised learning, SimCLR, BYOL, MAE, DINOv2

1. Introduction

Semantic segmentation assigns a class label to every pixel and is well suited to delineating a breast lesion as a coherent region. The project was designed as a two-stage study. Part A established the supervised benchmark and identified the strongest segmentation design for the dataset, BCSD-2024. Part B retained the same dataset partition and decoder design while replacing full supervision of the encoder with self-supervised pretraining. The main objective of Part B was to examine label efficiency: how much of the fully supervised performance could be recovered when only a smaller labelled subset was available for downstream segmentation.

Part A benchmarked DeepLabV3-ResNet50, SegFormer-B0 and YOLOv26-Sem under a common patient-level split. SegFormer-B0 was the strongest model and was therefore selected as the decoder/head design for the four Part-B experiments using SimCLR, BYOL, MAE and DINOv2. This setup allowed the SSL methods to be compared under the same dataset and downstream segmentation conditions while using substantially fewer labelled images.

2. Dataset and Leakage-Safe Split

The dataset is named BCSD-2024 breast ultrasound lesion segmentation. The dataset contains 1,521 image-mask pairs from 10 patients. Masks were treated as binary by thresholding grayscale values at 127, giving background (0) and lesion (1). A sampled scan of 80 masks found 3,797,319 foreground pixels

(1.00%) and 376,566,041 background pixels (99.00%), demonstrating severe pixel-level imbalance.

Split	Part-A role	Part-B role	Images
Train	Supervised training	Unlabelled SSL pretraining	1,035
Validation	Model validation	Labelled fine-tuning	318
Test	Held-out evaluation	Monitoring + final evaluation	168

Table 1. Final patient-level split inherited unchanged from Part A and remapped for Part B.

The final saved split was performed at patient level with seed 42: seven patients for training, two for validation, and one for test. Assertions verified disjoint patient sets, accounting for all 10 patients. This design prevents correlated ultrasound frames from the same patient appearing in multiple roles.

EDA also found non-uniform image dimensions: the sampled images ranged from 1664–1996 pixels in width and 2048–2457 pixels in height, with two unique widths and two unique heights. No corrupted/unreadable files were found. Two image-mask shape mismatches were detected; the downstream loaders handle such cases by resizing the mask with nearest-neighbour interpolation to the corresponding image dimensions before augmentation.

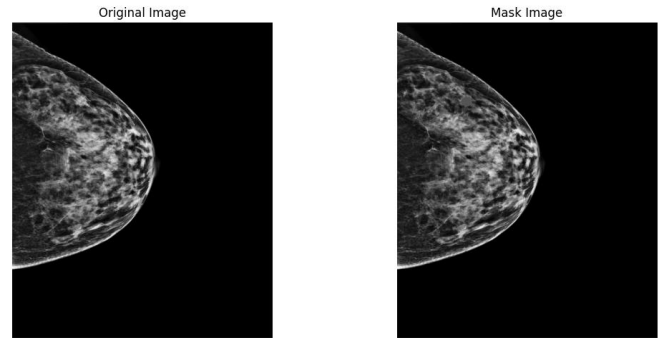


Fig. 1. Representative original ultrasound image and corresponding mask from the Part-A data preparation.

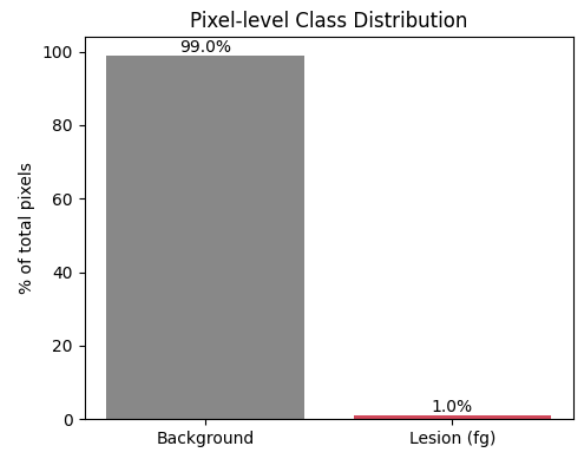


Fig. 2. Pixel-level class distribution from the Part-A EDA (80-mask sample).

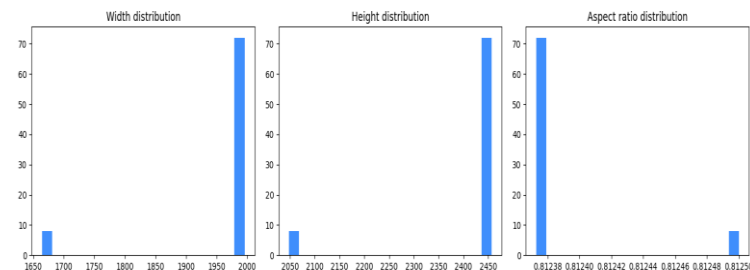


Fig. 3. Image-width, height, and aspect-ratio distributions from the Part-A EDA.

3. Part A: Supervised Benchmark

3.1 Data augmentation and preprocessing

All supervised models used 512×512 inputs. Training augmentation consisted of resize; horizontal flip ($p=0.5$); vertical flip ($p=0.3$); RandomRotate90 ($p=0.3$); ShiftScaleRotate with 5% translation, $\pm 10\%$ scale and $\pm 15^\circ$ rotation ($p=0.4$); RandomBrightnessContrast ($p=0.4$); GaussianBlur ($p=0.2$); and GaussNoise ($p=0.2$). Validation and test used deterministic resize only. Albumentations synchronized geometric image-mask transforms, while photometric transforms were applied to the image only.

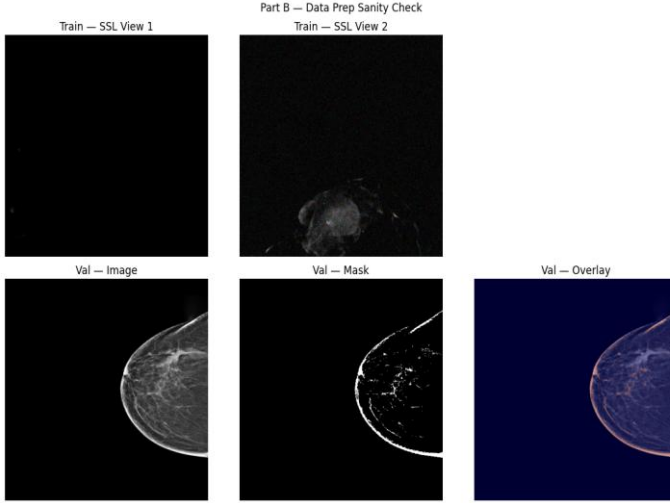


Fig. 4. Part-B sanity check: two augmented unlabeled views plus labelled image, mask, and overlay.

3.2 Model configurations

Model	Backbone/chec kpoint	Optimize r	LR	Epochs
DeepLabV 3	ResNet-50, COCO-pretrained	AdamW	1e-4	50
SegFormer -B0	nvidia/segforme r-b0-finetuned-ade-512-512	AdamW	6e-5	50
YOLOv26 -Sem	YOLOv26n-Sem, Cityscapes-pretrained	AdamW	1e-3	50

Table 2. Part-A core training configurations. [All runs used batch size 8 and seed 42]

DeepLabV3 and SegFormer-B0 used weighted cross-entropy plus Dice loss, with 0.5 weight on each component; DeepLabV3 also included an auxiliary loss term weighted by 0.4. SegFormer-B0 used five-epoch linear warmup followed by cosine annealing. YOLOv26-Sem used its Ultralytics semantic-segmentation loss. The class-weight vector used in the custom PyTorch models was approximately $[0.16, 1.84]$, strongly emphasizing the lesion class because of the 1% lesion-pixel prevalence.

3.3 Quantitative results

Model	mIoU	BG	Lesion	Pix Acc	Mean Acc	Dice
SegForm er-B0	0.8758	0.9938	0.7579	0.9939	0.9664	0.9296
DeepLab V3	0.8520	0.9924	0.7116	0.9925	0.9500	0.9138
YOLOv2 6-Sem	0.8402	0.9927	0.6877	0.9928	0.8778	0.9057

Table 3. Part-A test-set metrics on the common 168-image patient-level test set.

SegFormer-B0 achieved the highest mIoU and mean Dice, followed by DeepLabV3 and YOLOv26-Sem. Because the lesion occupies only about 1% of sampled pixels, the very high

background IoU and pixel accuracy should not be interpreted as equivalent foreground performance; lesion IoU and mIoU are more informative for the segmentation target.

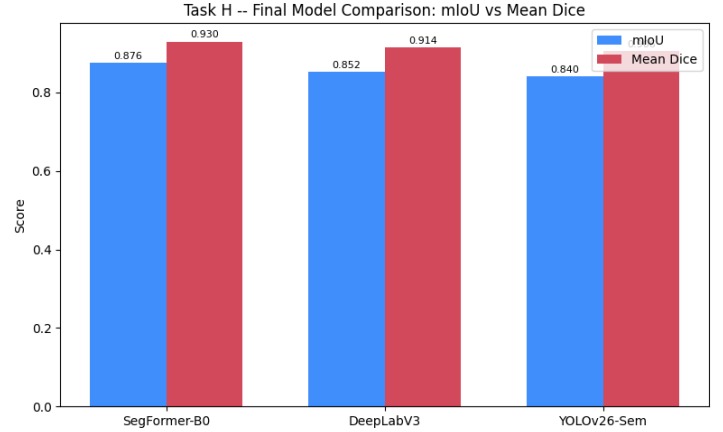


Fig. 5. Part-A final model comparison.

3.4 Part-A Error analysis

The error analysis shows that the models struggled with a few specific and consistently difficult images rather than making errors randomly. Several of the same difficult images appeared among the worst predictions, especially images from the P10 sequence, including the IMG-0007 and IMG-0011 cases. This happened across different architectures which suggests that these images are genuinely difficult rather than a problem of only one model.

The main difficulty was the lesion class, while the background was segmented very well. The confusion matrices also show this clearly: the large majority of pixels were correctly classified as background, while the more important errors came from missed or incorrectly predicted lesion pixels. This is understandable as the dataset is highly imbalanced with the background making up almost all pixels and the lesion covering only a small region. As a result, a model can achieve very high pixel accuracy while still missing important parts of the lesion. SegFormer-B0 showed this clearly: its background IoU was extremely high but its lesion IoU was much lower.

The models also made mistakes around difficult lesion boundaries and small foreground regions. Since similar failures appeared in more than one architecture, the results suggest that the dataset itself contains challenging visual cases that cannot be solved simply by changing the model. Overall, the part-A errors were mainly related to accurately locating and segmenting the small lesion region rather than recognizing the background.

SegFormer-B0 -- Pixel-level Confusion Matrix (row-normalized)

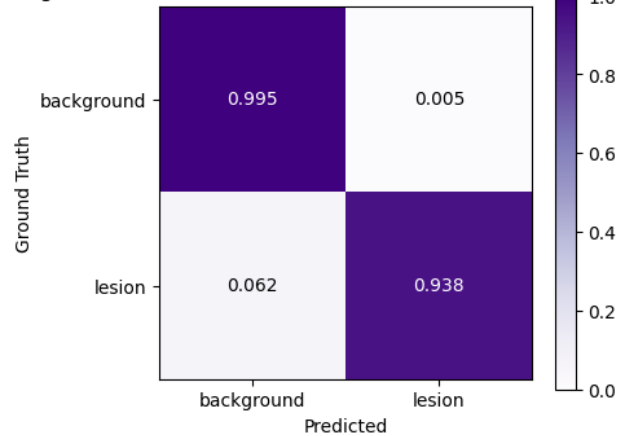


Fig. 6. SegFormer-B0 row-normalized pixel-level confusion matrix.

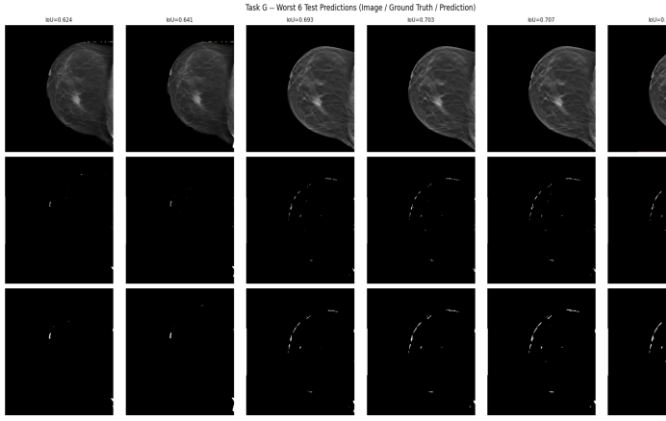


Fig. 7. SegFormer-B0 worst-case test predictions.

4. Part B: Self-Supervised Learning Pipeline

Part-B of the project reused the exact saved Part-A split. The 1,035 training images became the unlabelled SSL pretraining corpus and their masks were ignored during pretraining. The 318 validation images became the labelled fine-tuning set. The 168 test images were used both for downstream monitoring/checkpoint selection and final evaluation. This double role is an explicit assignment caveat: because the same test split influences checkpoint selection, reported final metrics can be somewhat optimistic relative to a truly untouched test set.

4.1 SSL methods and pretext tasks

Method	Encoder	Pretext task	Key configuration
SimCLR	ResNet-50	Contrastive NT-Xent	50 ep.; batch 8; T=0.5; AdamW 3e-4
BYOL	ResNet-50 online + target	Negative-free self-prediction	50 ep.; batch 2; grad accumulation 2; EMA 0.996; AdamW 3e-4
MAE	ViT-Base/16	Masked-patch reconstruction	50 ep.; batch 16; mask ratio 0.75; AdamW 1.5e-4
DINOv2	ViT-S/14	Student-teacher self-distillation	50 ep.; 2 global + 4 local crops; EMA 0.996; teacher T=0.04; student T=0.1

Table 4. Part-B self-supervised pretraining configurations.

SimCLR generated two independently augmented 224×224 views and optimized the NT-Xent objective with temperature 0.5. BYOL maintained online and EMA target networks without negative pairs. MAE continued pretraining of ViT-MAE at a 75% masking ratio. DINOv2 used two global 224×224 crops and four local 96×96 crops with an EMA teacher and centering.

4.2 Decoder attachment and adapters

All four SSL encoders were coupled to the SegFormer-style All-MLP decoder selected in Part A. CNN encoders passed their dense feature maps through 1×1 adapters producing channels [64, 128, 320, 512]. For ViT encoders, patch tokens were reshaped from sequences into 2-D feature grids, projected through the same adapter widths, decoded, and bilinearly upsampled. The token-to-grid adapter is necessary because the decoder expects spatial feature maps while transformer encoders produce patch-token sequences.

4.3 Downstream fine-tuning

Each SSL model was fine-tuned for exactly 50 epochs using the 318 labelled validation images. The reported runs used fully fine-tuned encoders (frozen_encoder=False), with AdamW, encoder learning rate $1e-5$, decoder/adaptor learning rate $1e-4$, cosine annealing, 224×224 input, and 0.5 weighted cross-entropy + 0.5 Dice. Test mIoU and loss were recorded each epoch, and the best monitoring checkpoint was selected for final test reporting.

Method	Pretrain min	Fine-tune min	Best FT epoch	Test mIoU
SimCLR	18.96	21.21	29	0.7232
BYOL	48.79	22.06	28	0.7102
MAE	22.05	15.48	19	0.4900
DINOv2	45.94	12.63	6	0.7633

Table 5. Part-B measured wall-clock times and best downstream checkpoint epochs

The measured SSL pretraining times were below 10 minutes per epoch for all four methods, so the project correctly kept one notebook per method rather than splitting pretraining and downstream stages. The timing requirement and the measured values are visible in the corresponding notebooks.

4.4 SSL pretext-loss evidence

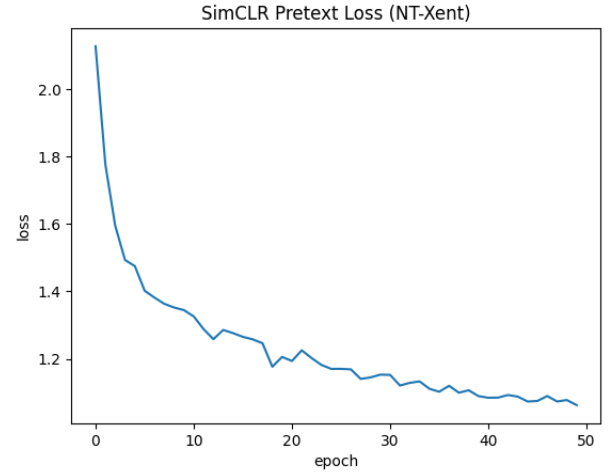


Fig. 8. SimCLR pretext loss curve

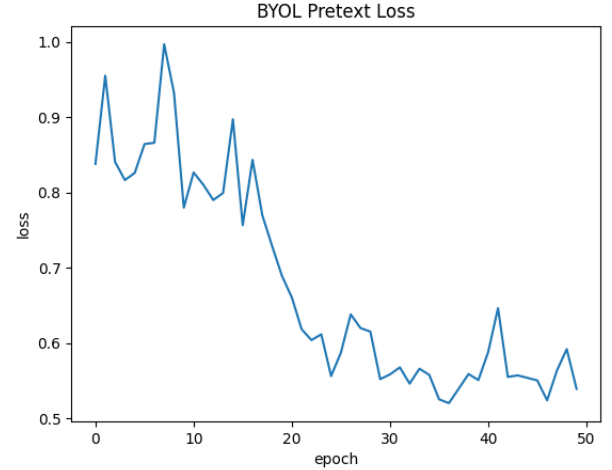


Fig. 9. BYOL pretext loss curve.

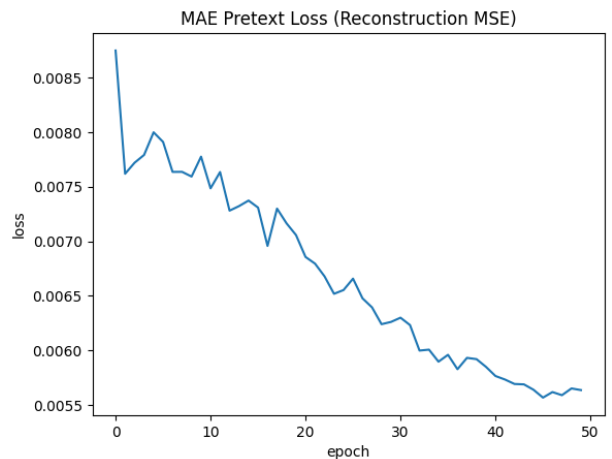


Fig. 10. MAE reconstruction-loss curve.

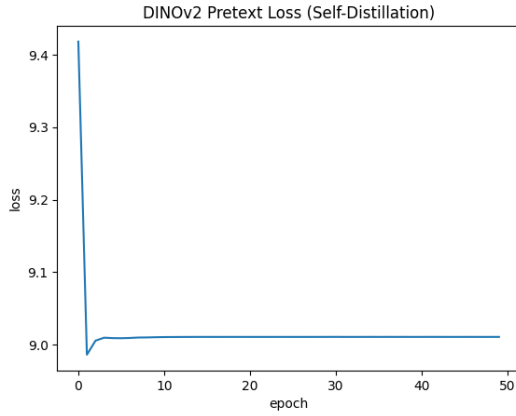


Fig. 11. DINOv2 distillation-loss curve.

4.5 SSL downstream training curves

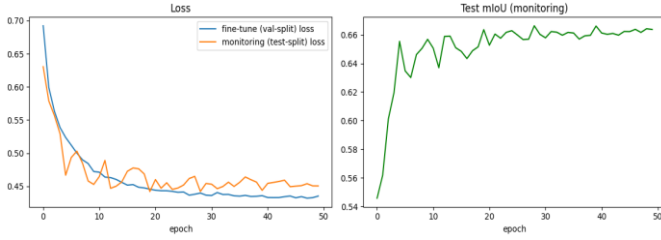


Fig. 12. SimCLR downstream training loss and test-mIoU monitoring curve.

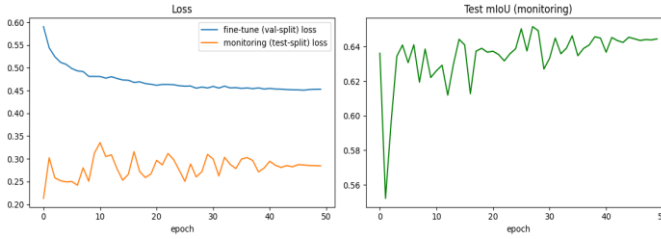


Fig. 13. BYOL downstream training loss and test-mIoU monitoring curve.

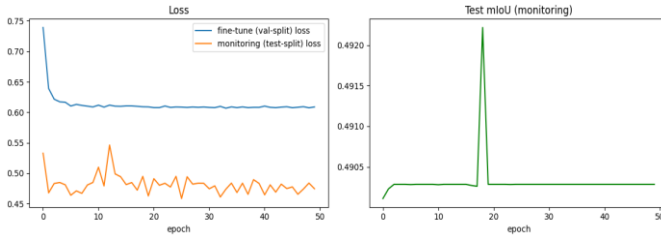


Fig. 14. MAE downstream training loss and test-mIoU monitoring curve.

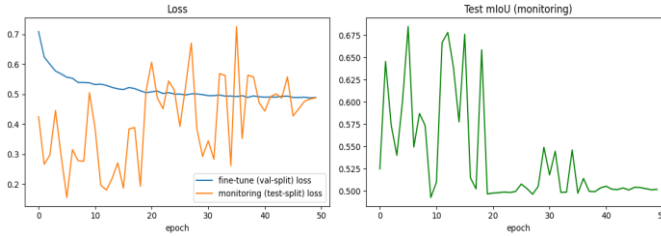


Fig. 15. DINOv2 downstream training loss and test-mIoU monitoring curve.

5. Results and Label-Efficiency Comparison

Method	mIoU	BG IoU	Lesion IoU	Mean Dice	Labelled	Label ratio	Recovered	Frozen
Part-A SegFormer-B0	0.8758	0.9938	0.7579	0.9296	1035	100%	100.0%	No
DINOv2	0.7633	0.9859	0.5408	0.8474	318	30.7%	87.2%	No
SimCLR	0.7232	0.9862	0.4602	0.8117	318	30.7%	82.6%	No
BYOL	0.7102	0.9852	0.4352	0.7995	318	30.7%	81.1%	No
MAE	0.4900	0.9798	0.0001	0.4950	318	30.7%	55.9%	No

Table 6. Consolidated Part-B comparison against the fully supervised Part-A baseline.

DINOv2 was the strongest label-efficient method, reaching mIoU 0.7633 with only 318 labelled images. This corresponds to 87.2% of the fully supervised SegFormer-B0 mIoU while using 30.7% of the labelled training-set size used by Part A. SimCLR and BYOL followed at 82.6% and 81.1% recovery. MAE was substantially weaker, primarily because its reported lesion IoU was almost zero (0.000129), indicating a failure to recover foreground segmentation despite a high background IoU.

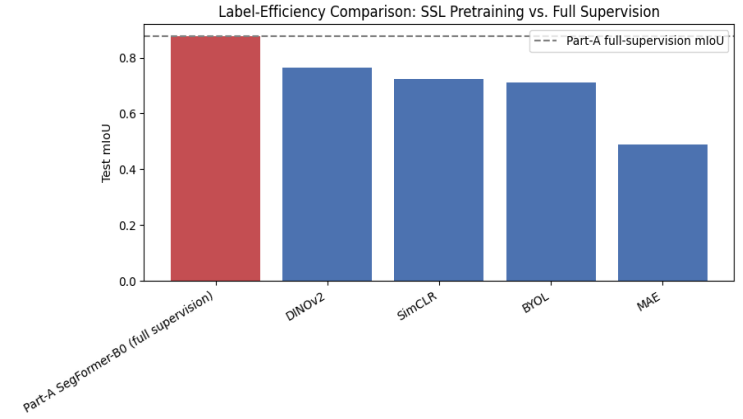


Fig. 16. Label-efficiency comparison from the final Part-B comparison notebook.

6. Error Analysis

The four worst-case prediction grids in Fig. 17-20 show that the SSL models did not struggle with completely different images. Several difficult cases were shared across methods, especially within the IMG-0007 sequence. In particular, IMG-0007-00012 and IMG-0007-00014 appeared among the worst six cases for SimCLR, BYOL and DINOv2, while IMG-0007-00013, IMG-0007-00010 and IMG-0007-00015 were also difficult for more than one method. Seeing the same images repeatedly in the worst-case grids indicates that some ultrasound images are simply harder to segment rather than the errors being caused only by a particular SSL model.

The class-wise results show a similar pattern to Part-A: the background was segmented much more easily than the lesion. DINOv2 handled the lesion better than the other SSL models, while SimCLR and BYOL were weaker. MAE showed the clearest failure, producing almost no meaningful lesion segmentation even though its background prediction remained relatively strong.

Looking at both the worst-case predictions and the quantitative results, the main challenge in Part-B was not recognizing the overall image structure but transferring the learned features into accurate pixel-level localization of the small lesion region. The fact that several SSL methods struggled with the same images seen in Part-A also indicates that self-supervised pretraining could not completely overcome the underlying visual difficulty of those cases. The error analysis highlights that foreground lesion segmentation was much harder than background segmentation.

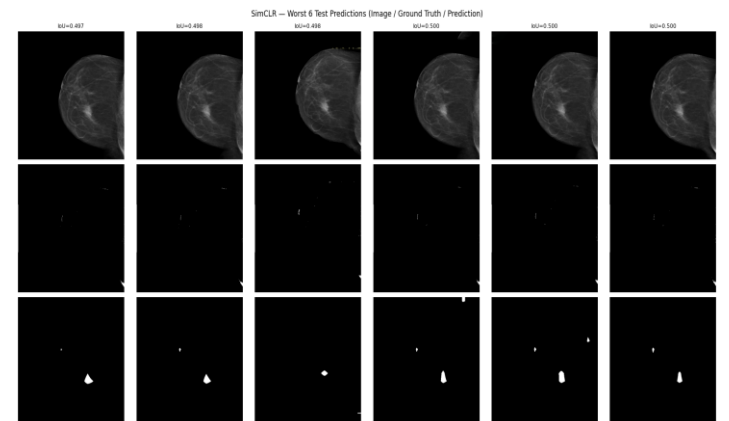


Fig. 17. SimCLR worst-six test predictions.

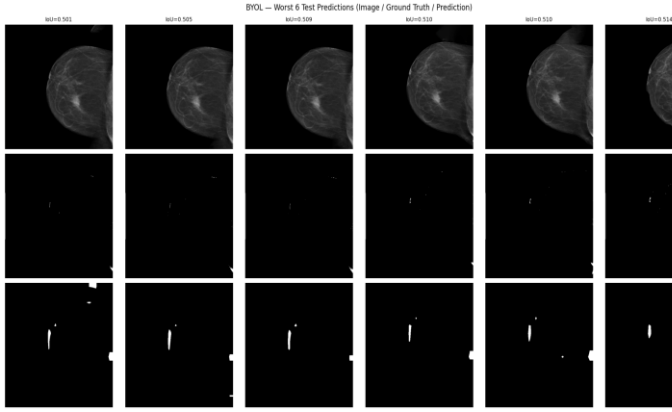


Fig. 18. BYOL worst-six test predictions.

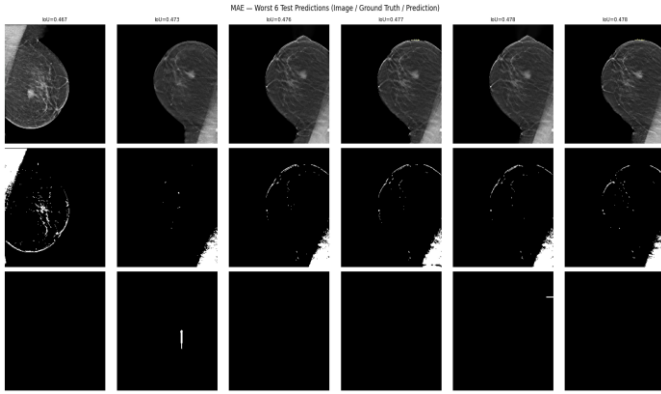


Fig. 19. MAE worst-six test predictions.

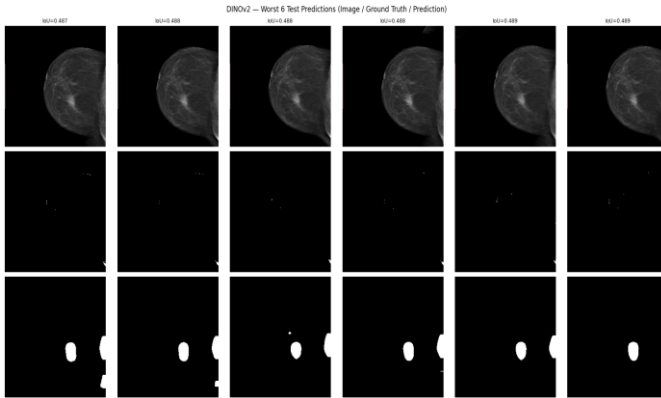


Fig. 20. DINOv2 worst-six test predictions.

The SSL confusion matrices show a common pattern of very high background agreement with much lower lesion recognition. This reinforces the importance of lesion IoU and mIoU for evaluating this highly imbalanced segmentation problem.

7. Insights

The main lesson from our project is that a self-supervised pretrained encoder does not automatically give good segmentation. What matters is how well the learned features transfer to the actual task. In our case, DINOv2 achieved the highest performance [SSL mIoU (0.7633) and lesion IoU (0.5408)] with SimCLR and BYOL next in line, whereas MAE fared significantly worse, despite all four techniques utilizing the same labeled validation set for fine-tuning. This indicates that the quality of the learned representation was a major factor.

DINOv2 was the most suitable for our dataset as its self-distillation features transferred more effectively to lesion segmentation. It managed to acquire valuable visual representations without employing lesion labels in the pretraining phase. SimCLR and BYOL proved to be useful but their features were not as effective for the dense pixel-level task.

The MAE outcome was the most unexpected. Despite the model finishing pretraining and fine-tuning, its lesion IoU remained

nearly zero. One probable explanation is that reconstructing masked patches helps the model in grasping the overall image structure, whereas our task requires accurately locating a relatively small lesion region. The severe foreground imbalance exacerbated the situation: the background takes up most pixels, allowing the models to excel in background prediction while still neglecting crucial lesion regions.

The experiment comparing frozen and fine-tuned SimCLR indicated that fine-tuning the encoder outperformed leaving it frozen. This means the pretrained features were beneficial, yet they required adjustment for the particular segmentation task.

Another key aspect is label efficiency. The fully supervised Part-A model [SegFormer-B0] remained the strongest overall, but DINOv2 achieved a significant portion of that performance utilizing far fewer labelled images. This demonstrates the main practical value of SSL in our project: minimizing the amount of labelled data needed instead of completely substituting full supervision.

DINOv2 also selected an early checkpoint during downstream monitoring. This highlights that useful segmentation features were learned relatively quickly and continuing training was not necessarily useful on the monitored split.

With more computational resources, the weak MAE performance could be further explored through experiments with various masking ratios, patch sizes, pretrained checkpoints and extended pretraining durations. Enhanced input resolution might also be investigated to more effectively retrain the small lesion details. A more systematic comparison between frozen and fine-tuned settings across the SSL methods, along with various decoder/adaptor configurations for MAE, could provide further insight into the observed performance gap.

8. Conclusion

This project established a supervised benchmark on the BCSD-2024 dataset and then examined how self-supervised pretraining can reduce the amount of labelled data required for semantic segmentation. In Part A, SegFormer-B0 achieved the best fully supervised performance, with a test mIoU of 0.8758. Building on this result, Part B evaluated four SSL approaches using only 318 labelled images for downstream fine-tuning. DINOv2 achieved the strongest SSL performance, reaching a test mIoU of 0.7633, which corresponds to 87.2% of the Part-A baseline while using only 30.7% of its labelled data. SimCLR and BYOL also retained a substantial portion of the supervised performance, whereas the tested MAE configuration struggled particularly with lesion segmentation. Finally, the results show that self-supervised pretraining, especially DINOv2 in this study, can provide a practical way to achieve strong segmentation performance when labelled medical images are limited.

9. Notebook Links

The completed project contains 10 notebooks [one notebook per method].

Part-A

NB0 — EDA & Data Prep:

<https://www.kaggle.com/code/sumaiyahoque/nb-0-eda-and-data-prep/notebook>

NB1 — DeepLabV3:

<https://www.kaggle.com/code/sumaiyahoque/nb-1-seg-deeplabv3>

NB2 — SegFormer-B0:

<https://www.kaggle.com/code/sumaiyahoque/nb-02-seg-segformer-b0>

NB3 — YOLOv26-Sem + Part-A's model comparison:

<https://www.kaggle.com/code/sumaiyahoque/nb-03-seg-yolov26-semantic>

Part-B

NB4 — Part-B Data Prep:

<https://www.kaggle.com/code/sumaiyahoque/nb-04-partb-data-prep>

NB5 — SimCLR:

<https://www.kaggle.com/code/sumaiyahoque/nb-05-partb-simclr>

NB6 — BYOL:

<https://www.kaggle.com/code/sumaiyahoqueriyan/nb-06-partb-byol>

NB7 — MAE: <https://www.kaggle.com/code/sumaiyahoque/nb-07-partb-mae>

NB8 — DINOv2:

<https://www.kaggle.com/code/sumaiyahoque/nb-08-partb-dinov2/notebook>

NB9 — Final Comparison:

<https://www.kaggle.com/code/sumaiyahoque/nb-09-partb-final-comparison>

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